

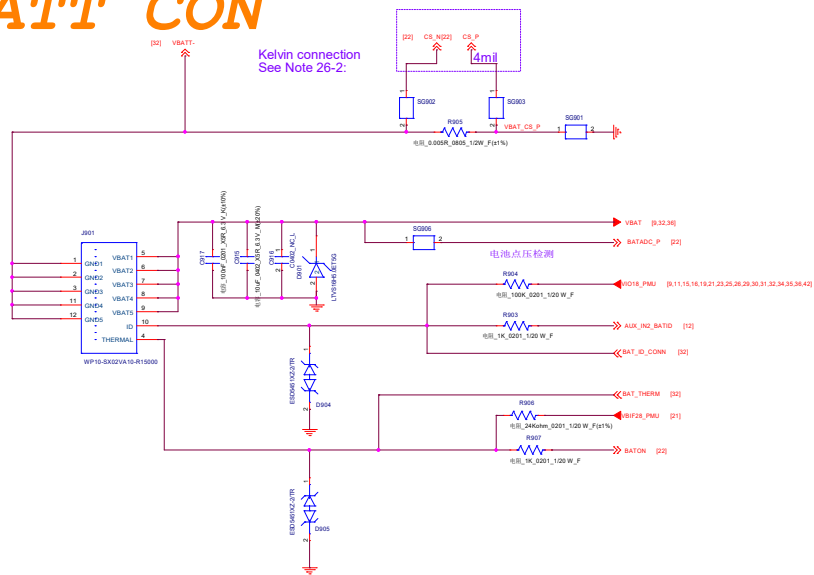
SN-A145P

I2C Configuration Table

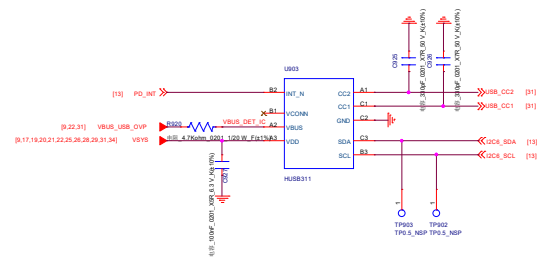
Function	I2C Slave Address (7-bit mode)
Page 15 : Charger IC	
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Page 23 : Camera	
E-COMPASS	
ALS&PS SENSOR	
NFC	

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- 05.BB_POWER_IO
- 06.BB_1_GPIO
- 07.BB_2_GPIO
- 08.BB_3_EMI
- 09.BB_AUXADC
- 10.POWER_MT6357_Buck
- 11.POWER_MT6357_LDO
- 12.POWER_MT6357_IF
- 13.POWER_MT6357_Audio
- 14.POWER_BAT_CON
- 15.CHARGE_IC+CC
- 16.LCD BACKLIGHT+LCD BIAS
- 17.FLASHLED DRIVER
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- 24.CAMERA
- 25.LINK
- 26.SIM_SD
- 27.SideKey_Flash_LED
- 28.Sensor
- 29.Test Point_JTAG
- 30.RF_MT6177_Pin_Out
- 31.RF_MT6177_RF_TX
- 32.RF_MT6177_RF_PRX
- 33.RF_MT6177_RF_DRX
- 34.RF_ANT
- 35.WCN_FEM
- 36.CONNECTIVITY_CONSYS_MT6631
- 37.SAR_Sensor

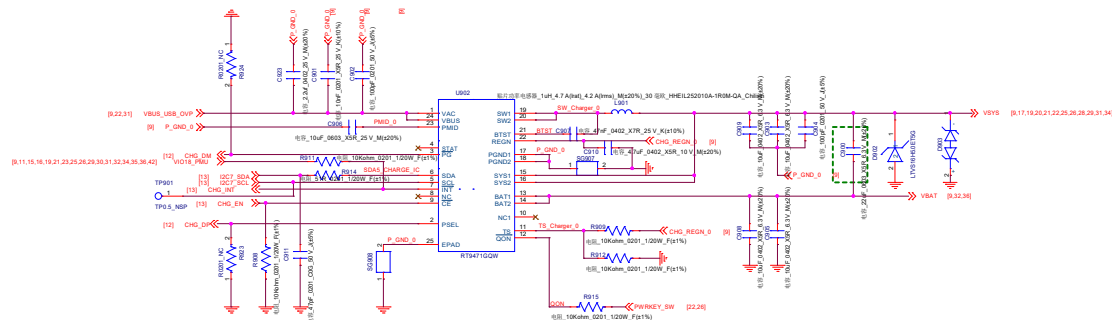
BATT CON



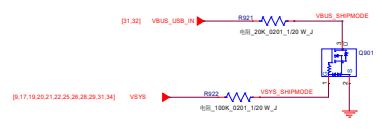
PD

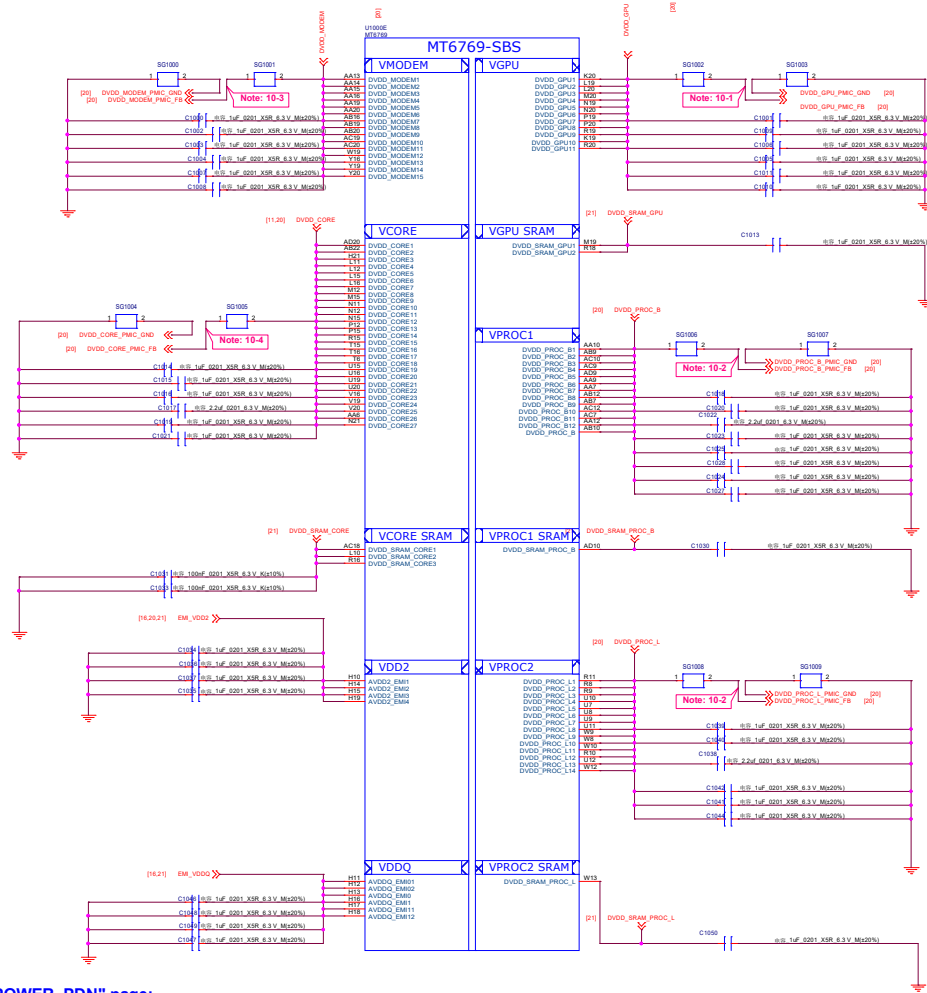


Charger



Ship Mode Det





Schematic design notice of "10_BB_POWER_PDN" page:

- Note 10-1:

Differential pair of DVDD_GPU remote sense must be close to BB's ball.

Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-2:

Differential pair of DVDD_PROC remote sense must be close to BB's ball.

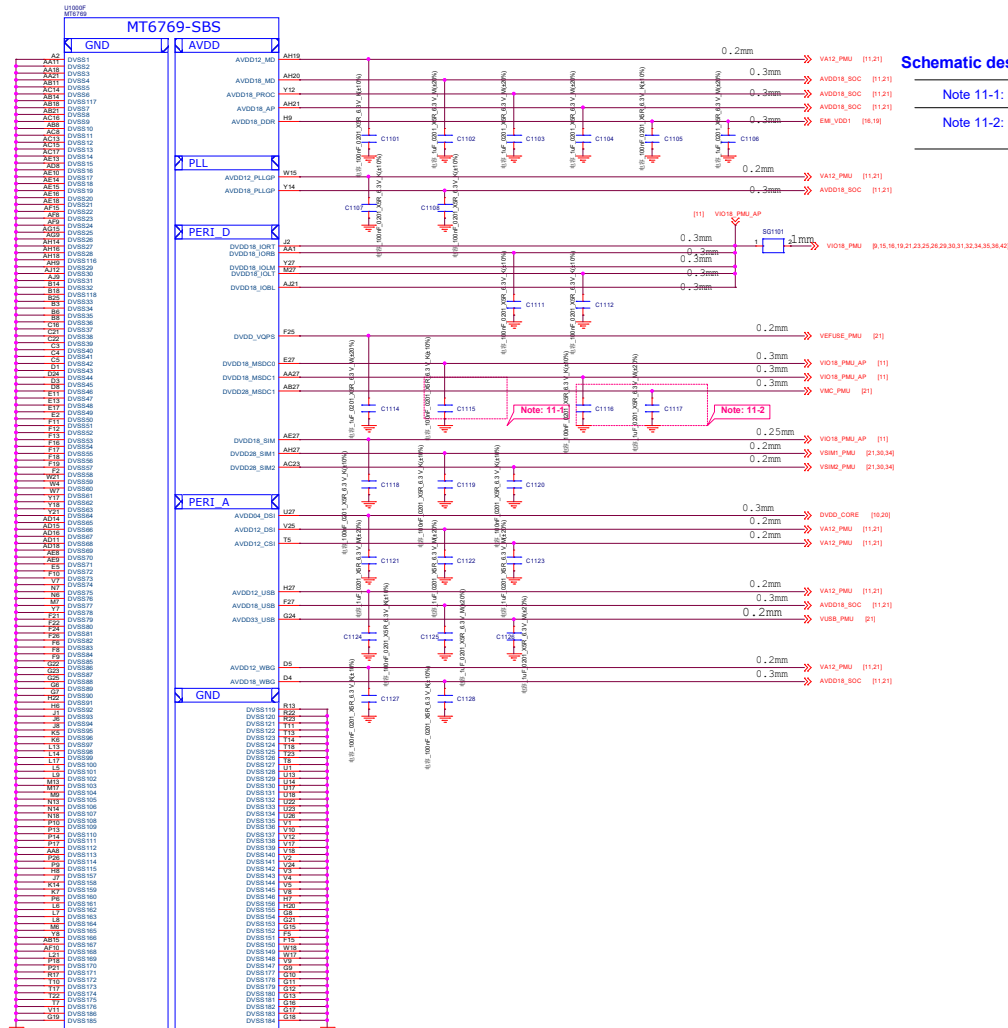
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-3:

Differential pair of DVDD_MODEM remote sense must be close to BB's ball.

Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-4:

Differential pair of DVDD_CORE remote sense must be close to BB's ball.

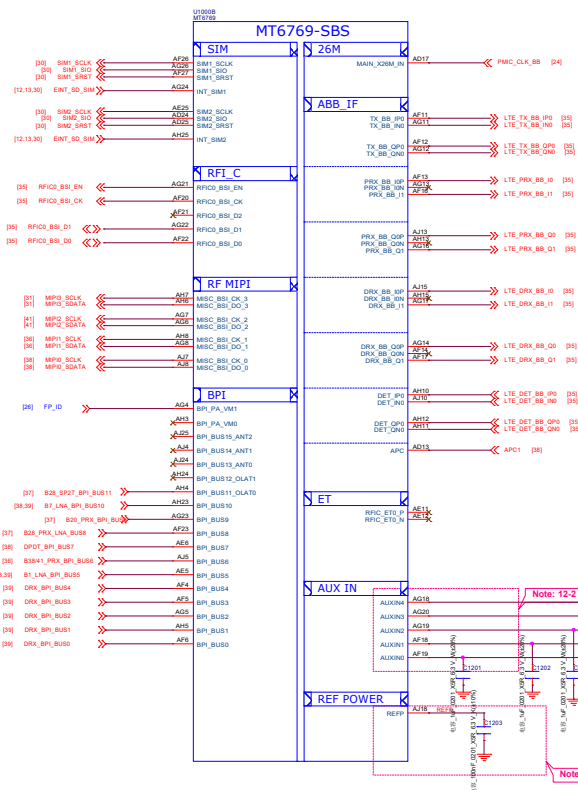
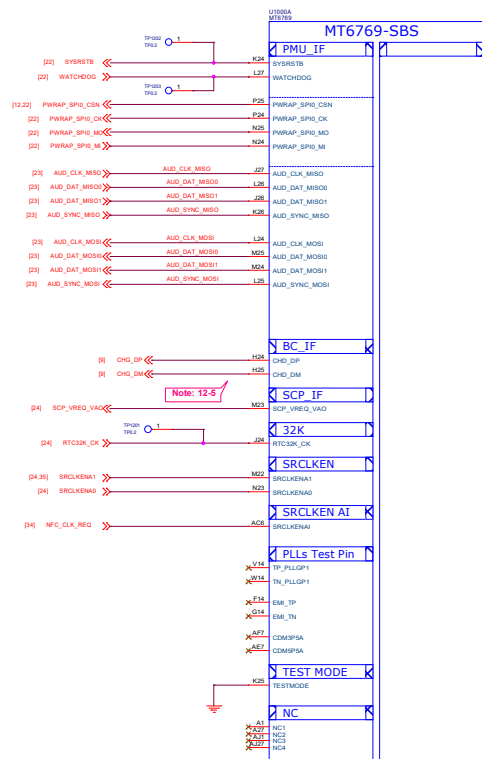
Remote sense trace with GND shielding to PMIC (Differential)



Schematic design notice of "11_BB_POWER_IO" page:

Note 11-1: C1115 closed DVDD18_MSDC0 150mil

Note 11-2: C1116 closed DVDD18_MSDC1 150mil
C1117 closed DVDD28_MSDC1 150mil



Note: USB_THERM_NTC默认不使用, C1221脚0R, R3107 NC

Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
		AP_JTAG	MD_JTAG
default=PU	default=PD	N/A	N/A
HI	LO	N/A	N/A
HI	HI (by ext. PDI)	SPI0 + EINT8	SPI2 + SPI3
LO (by ext. PDI)	LO	SPI0 + EINT8	N/A
LO	HI	N/A	N/A

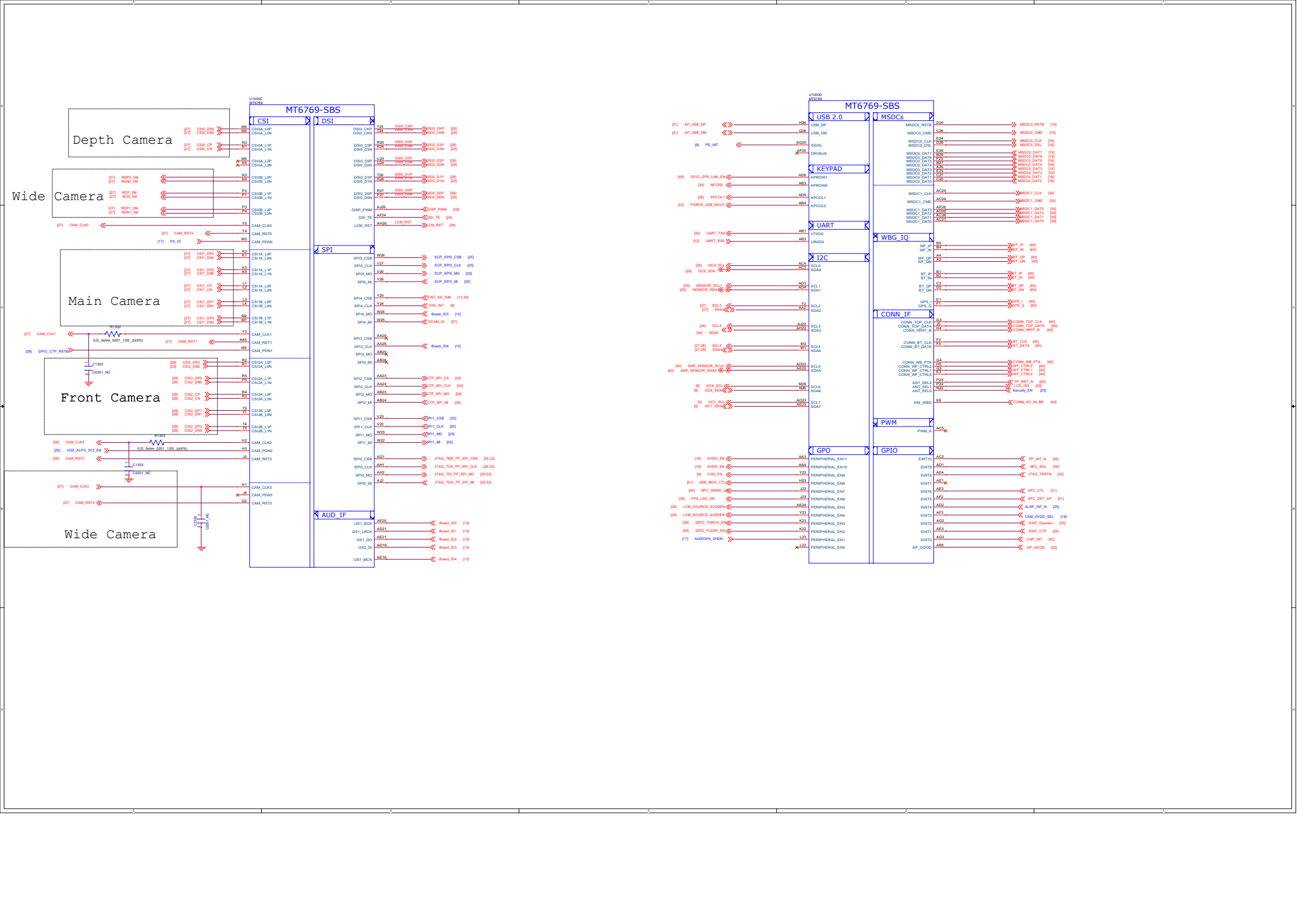
Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

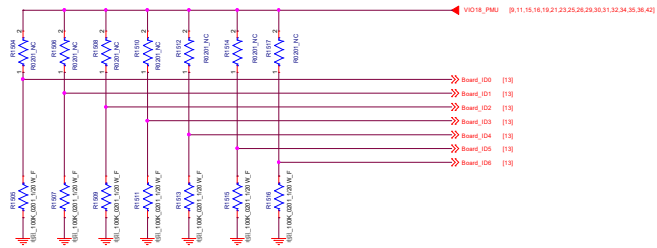
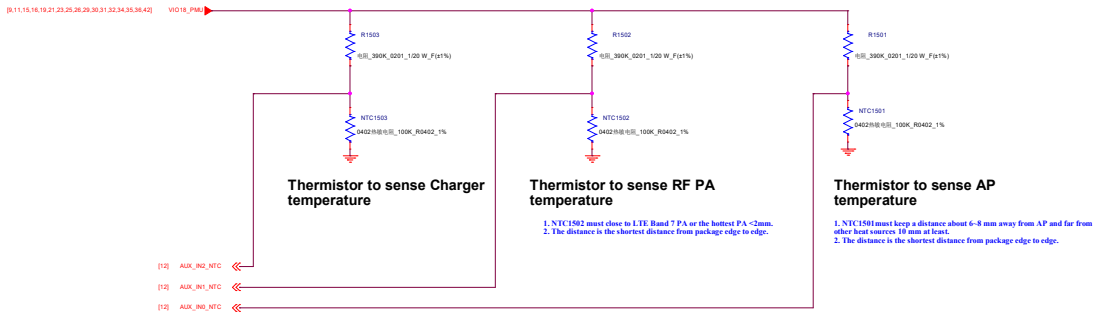
Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

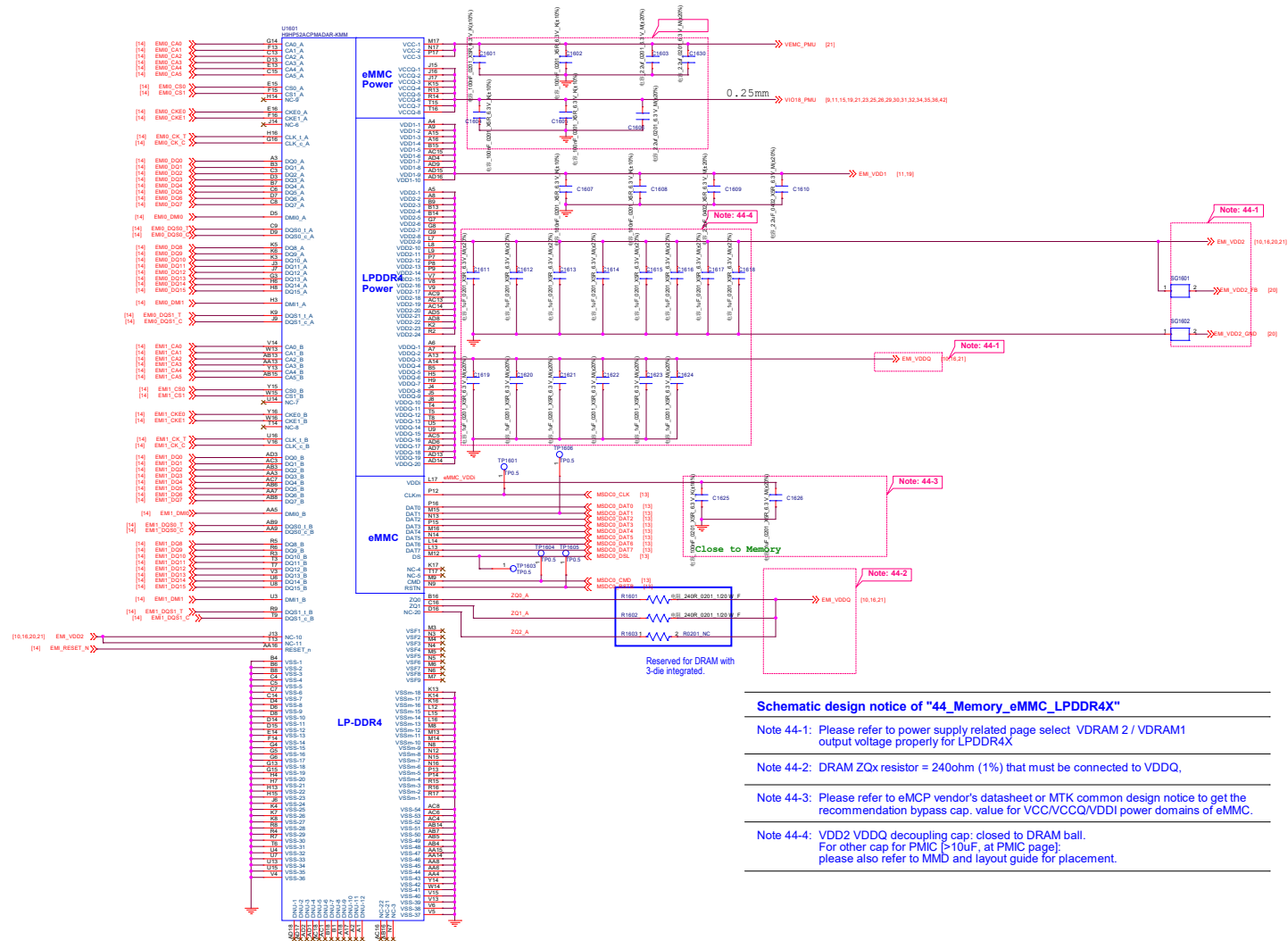
Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / 0.7v	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved

Note 12-5: Charger must have D+/D- pin for charger type USB detection.
Charger must have at least 500mA USB current for All charger type.







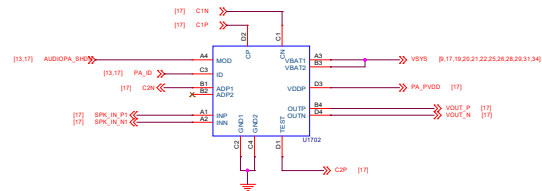
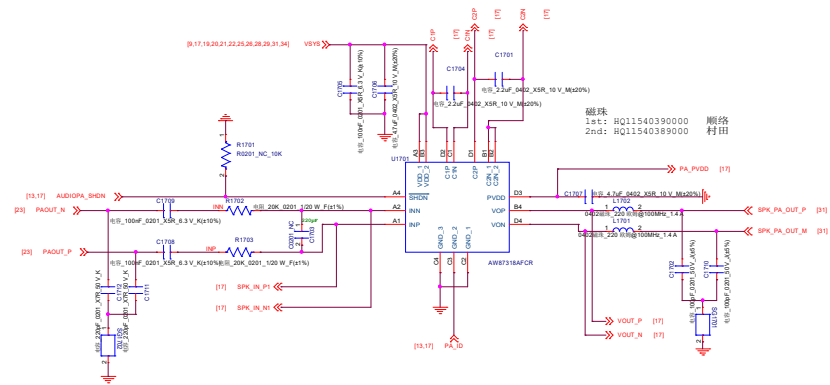
Schematic design notice of "44_Memory_eMMC_LPDDR4X"

Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

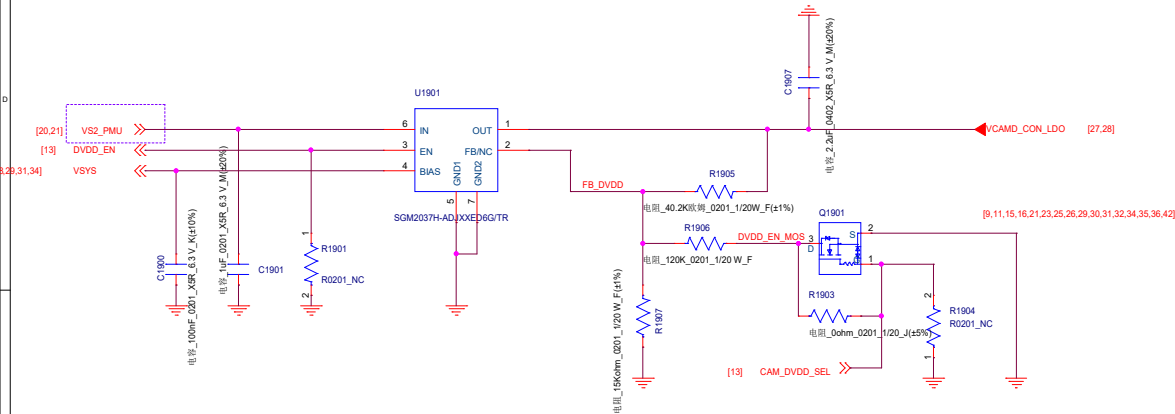
Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [~10uF, at PMIC page], please also refer to MMD and layout guide for placement.

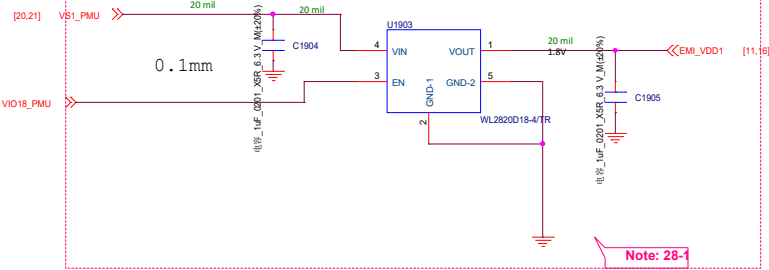


LDO forUM 5M(VCAMD_EXT1)
1.2V

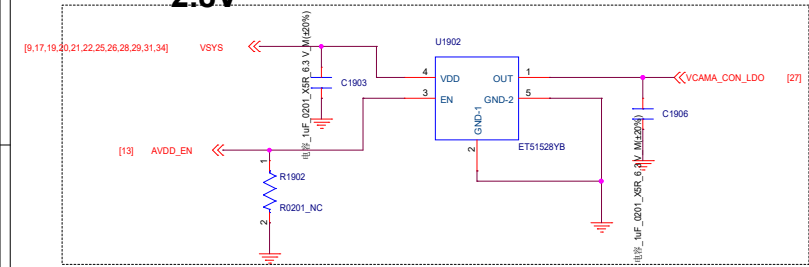


R1905	R1906	R1907	Q1901	R1903	R1904	电压	CAM_DVDD_EN
ON	ON	ON	NC	ON	NC	1.1V	Float
ON	ON	ON	NC	ON	NC	1.2V	L

Ext. EMI_VDD1 for LPDDR4X VDD1

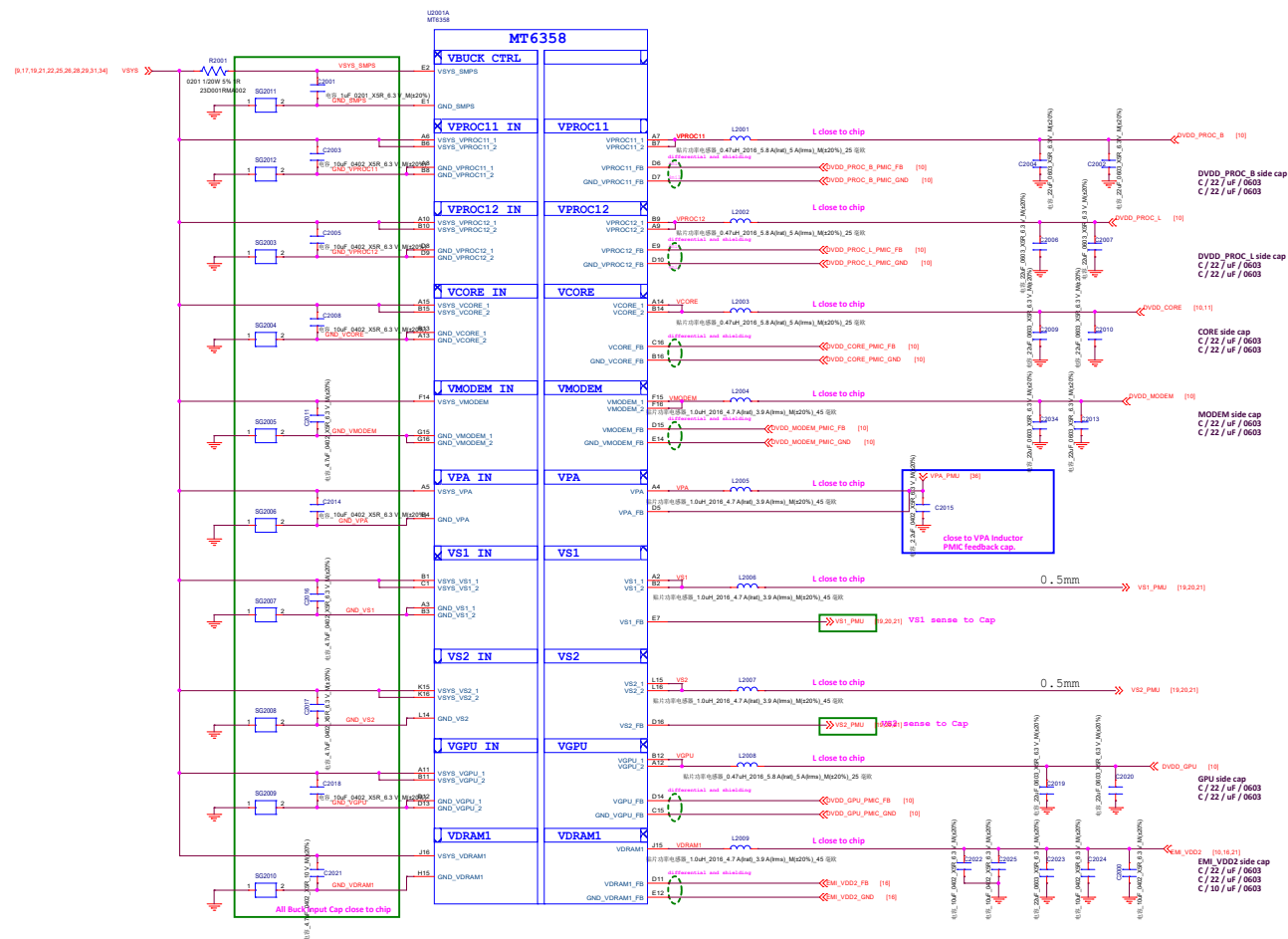


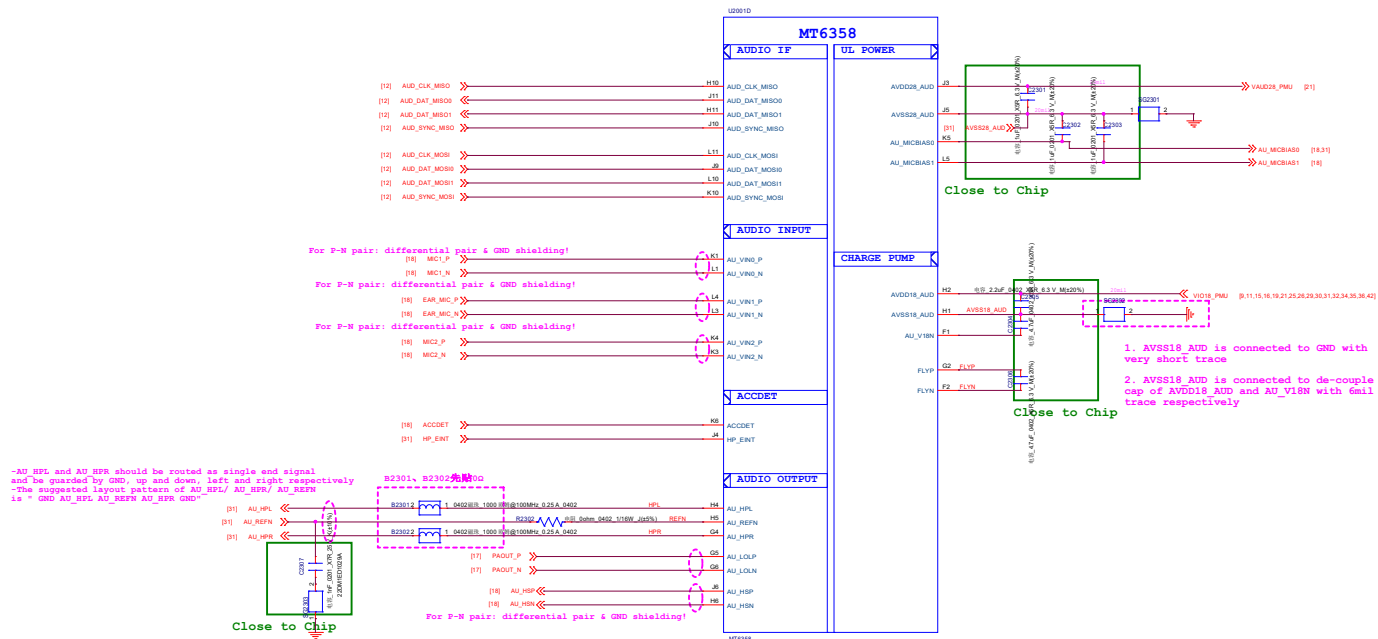
LDO for Dual Camera (VCAMA_EXT)
2.8V

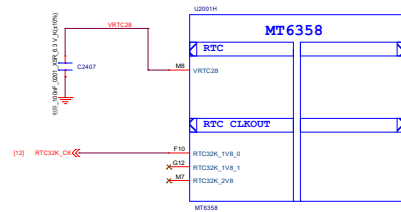
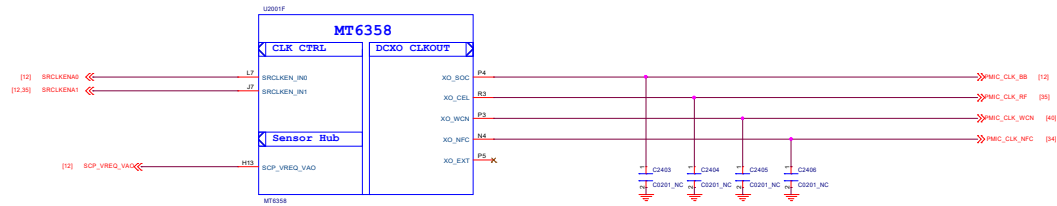
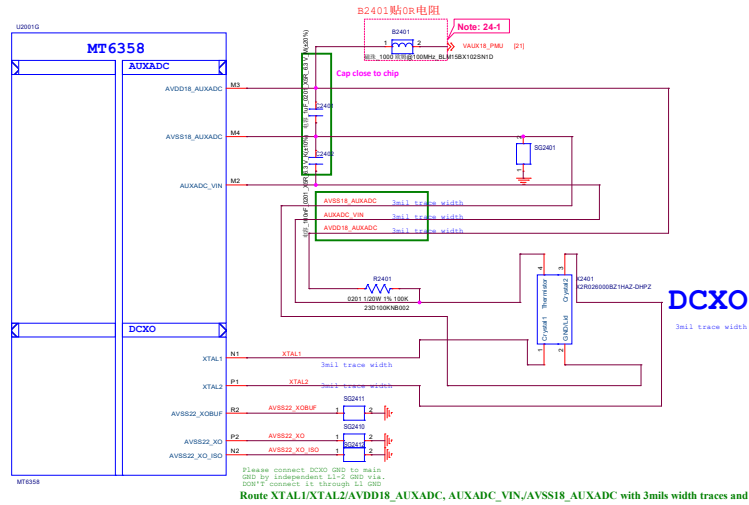


Schematic design notice of "28_POWER_ThirdParty_Powers" page:

Note 28-1: Refer to BB design notice for LP4X VDD1 LDO selection guide



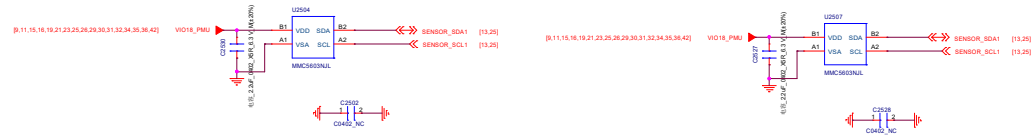




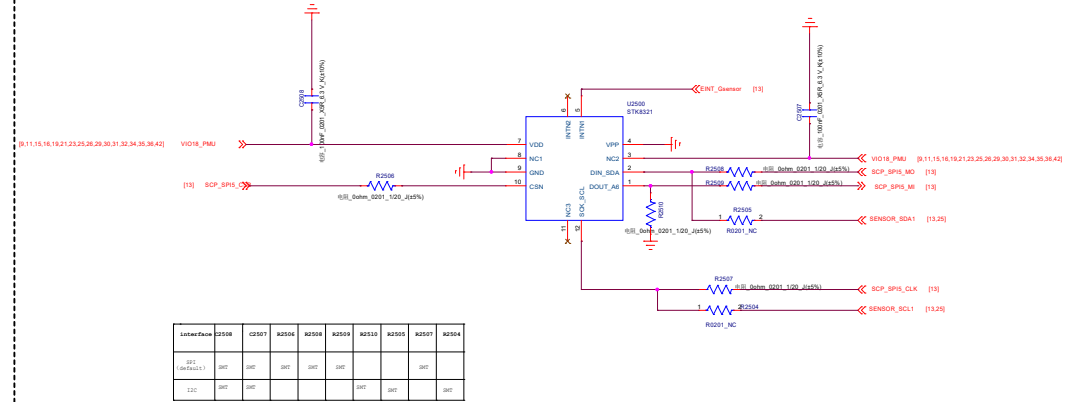
Schematic design notice of "24_POWER_MT6358_Clock"

Note 24-1: Please follow MT6768_MT6358 Co-Clock Design Notice for Layout guide of VAUX18, then B2401 can use 0 ohm to replace BEAD.

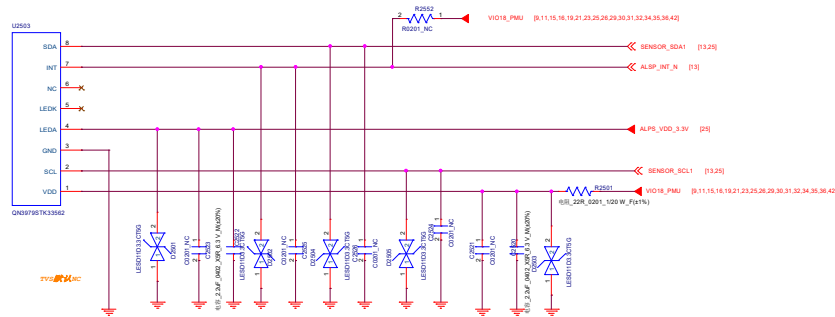
eCOMPASS



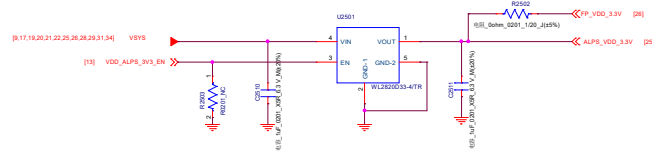
G SENSOR



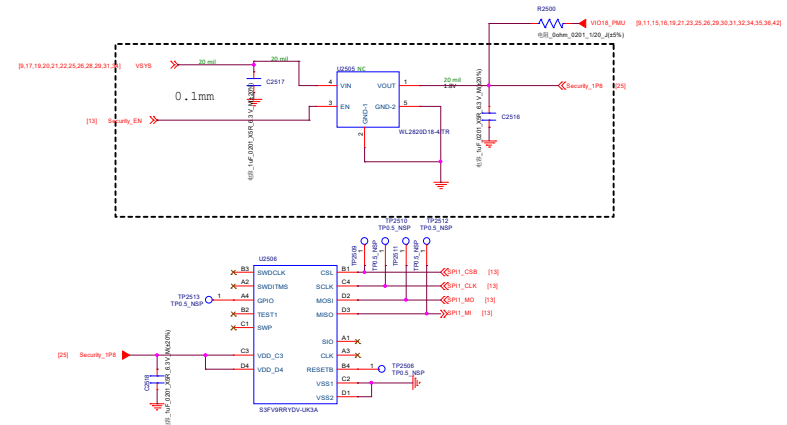
ALPS Sensor



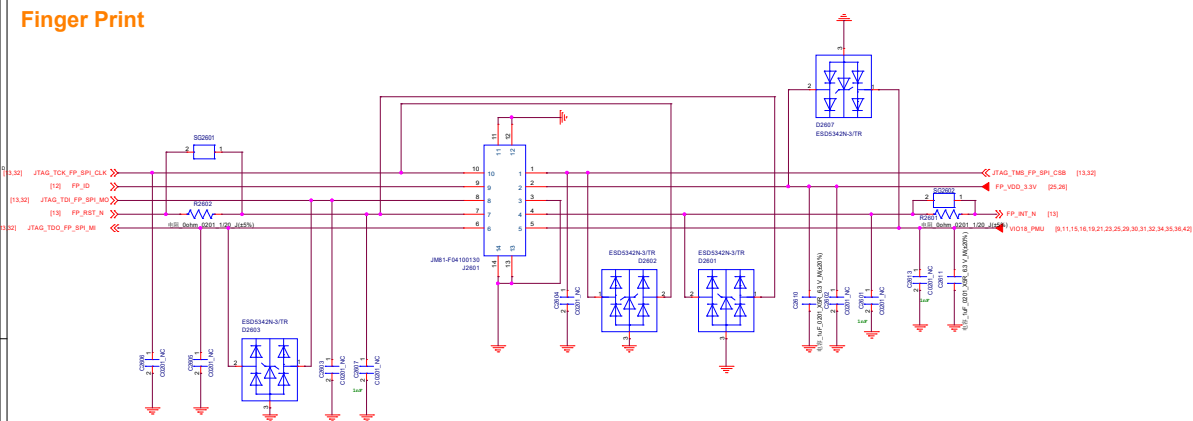
LDO For ALP



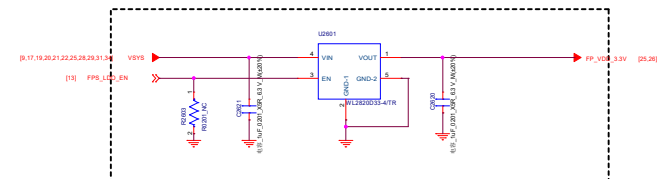
Security IC



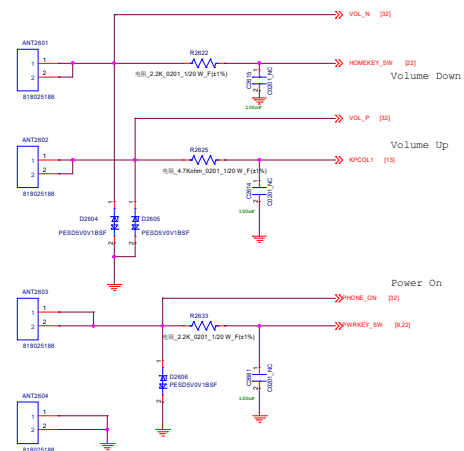
Finger Print



LDO for FPS

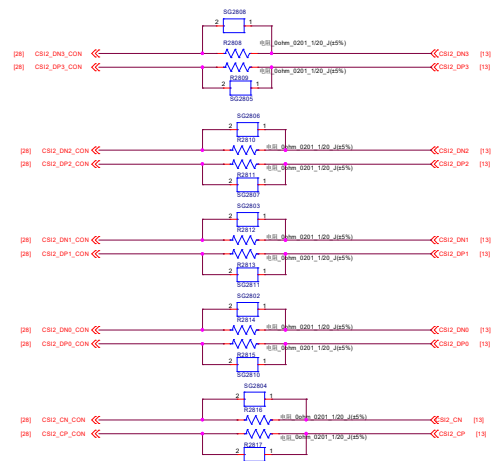
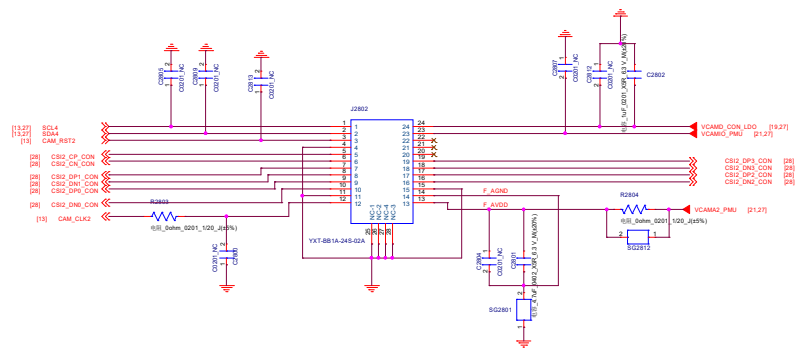


KEY pad

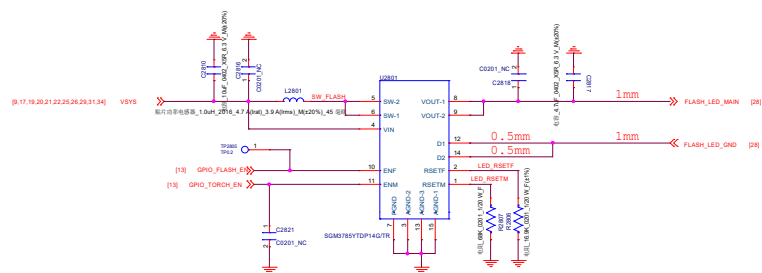


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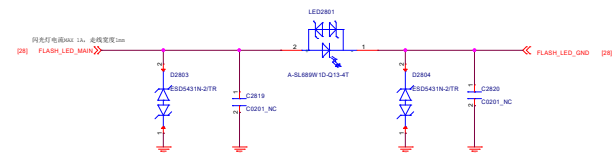
Front 13M Cam

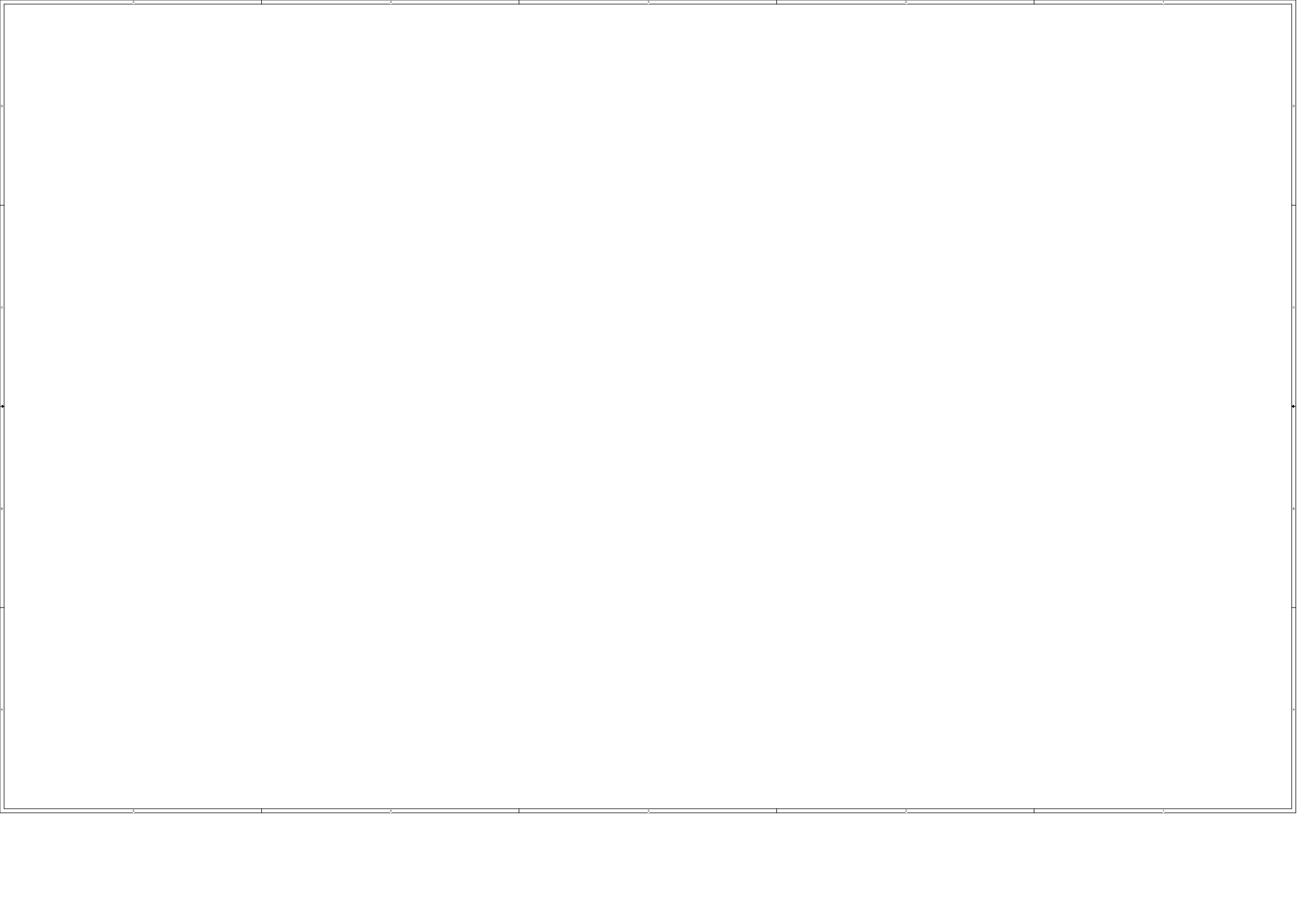


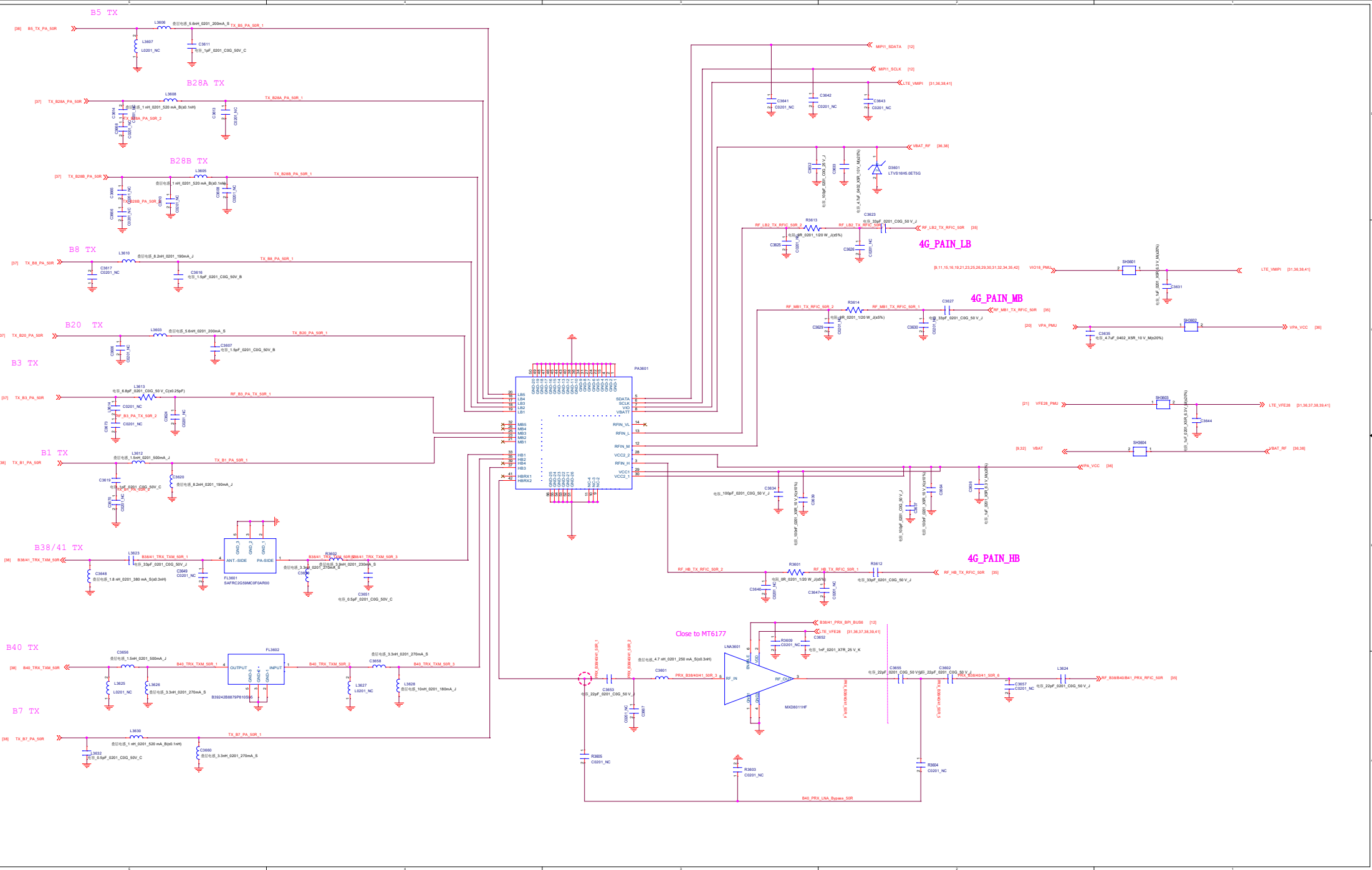
Flash Driver

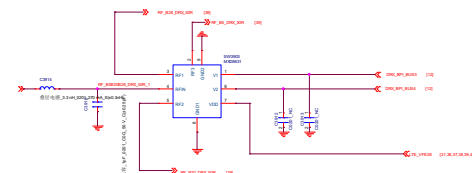
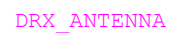


Flash LED

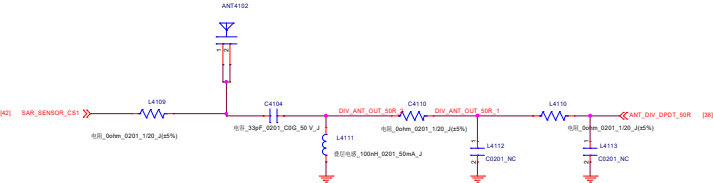




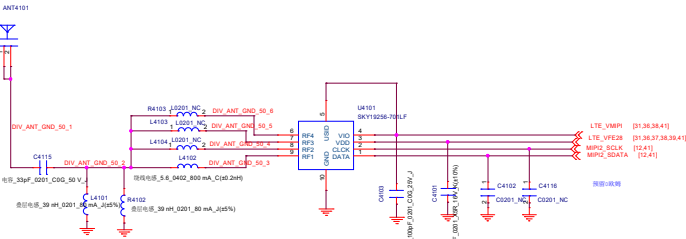




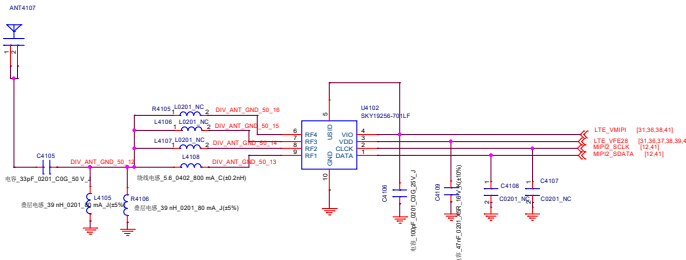
DIV ANT



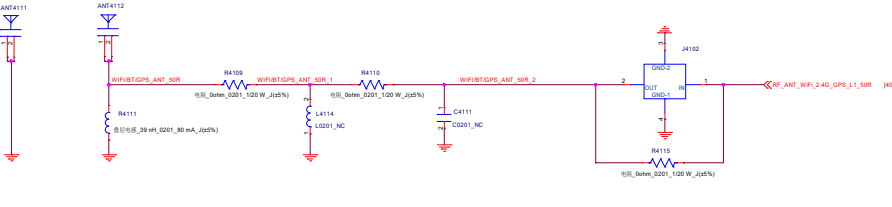
DIV GND



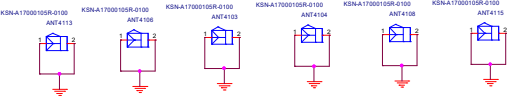
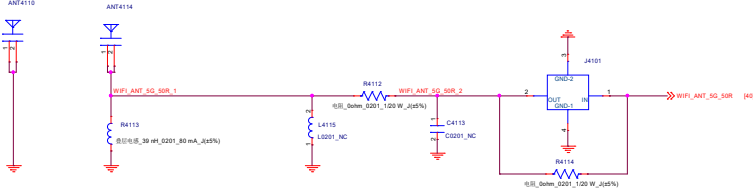
DIV GND

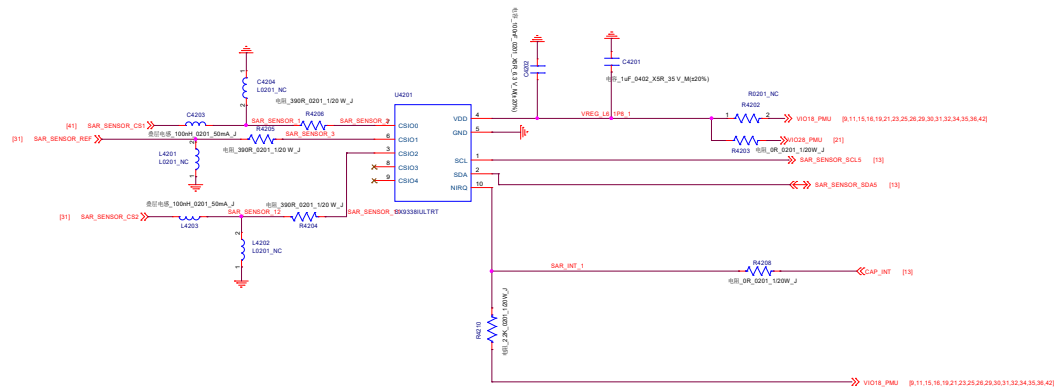


2.4G_GPS ANT

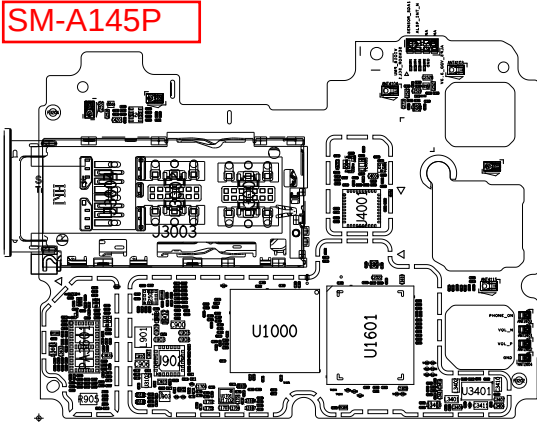


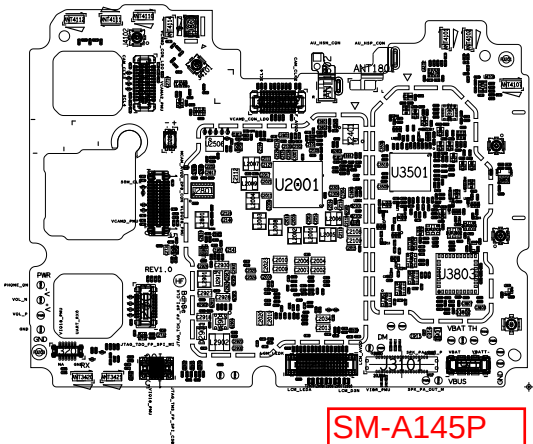
Wifi 5G ANT





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